

option 1

Excellent choice. **Option 1** will produce much better quality notes. I'll write them exactly as a university lecturer would teach in class, with explanations, derivations, board work, diagrams, examples, and questions.

Let's start with the **first topic: Virtual Temperature**.

ATMOSPHERIC PHYSICS

Lecture Note 1

Virtual Temperature

Course: FYUGP B.Sc. Physics

Duration: 40–50 minutes

Learning Outcomes

At the end of this lecture, students should be able to:

- Define virtual temperature.
 - Explain why moist air is lighter than dry air.
 - Derive the expression for virtual temperature.
 - Calculate virtual temperature.
 - Explain its importance in meteorology.
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Introduction

Ask students the following question:

Which is heavier? Dry air or moist air?

Most students will answer:

Moist air

Then say,

"Surprisingly, the correct answer is **dry air**."

This surprises students and immediately gets their attention.

Why is Moist Air Lighter?

Air is a mixture of gases.

Gas	Percentage
Nitrogen	78%
Oxygen	21%
Argon	0.93%
Carbon dioxide	0.04%
Water vapour	Variable

Water vapour replaces some nitrogen and oxygen molecules.

Compare molecular masses:

Nitrogen

$$N_2 = 28$$

Oxygen

$$O_2 = 32$$

Average molecular weight of dry air

Water vapour

$$H_2O = 18$$

Since

$$18 < 29$$

water vapour is lighter.

Therefore,

Moist air becomes less dense.

Classroom Demonstration

Ask students,

Which is lighter?

A balloon filled with hot air

or

A balloon filled with normal air?

Everyone knows

Hot air balloon rises.

Similarly,

Moist air also becomes lighter.

Density of Air

From the ideal gas equation,

$$PV = nRT$$

or

$$P = \rho RT$$

where

P = Pressure

ρ = Density

R = Gas constant

T = Temperature

Rearranging,

$$\rho = \frac{P}{RT}$$

Observe carefully.

If

Temperature increases,

Density decreases.

Similarly,

When moisture increases,

Density also decreases.

Why Do Meteorologists Use Virtual Temperature?

Suppose

Temperature

=30°C

One day

Humidity

20%

Another day

Humidity

95%

Are both air masses equally dense?

No.

The humid air is lighter.

Therefore,

Meteorologists define

Virtual Temperature.

Definition

Virtual Temperature is

The temperature that dry air must have to possess the same density as moist air at the same pressure.

Symbol

T_v

Unit

Kelvin (K)

Mathematical Expression

For atmospheric applications,

$$T_v = T(1 + 0.61r)$$

where

T = Actual temperature (K)

r = Mixing ratio (kg/kg)

Meaning of the Formula

If

Humidity increases

↓

Mixing ratio increases

↓

Virtual temperature increases

This does **not** necessarily mean the air is actually hotter.

It means

The moist air behaves **as if** it were warmer because it is less dense.

Physical Interpretation

Suppose

Actual temperature

300 K

Dry air

Density

1.18 kg m^{-3}

Moist air

Density

1.14 kg m^{-3}

The moist air behaves like dry air at a higher temperature.

That higher temperature is called

Virtual Temperature.

Numerical Example

Temperature

$$T = 300K$$

Mixing ratio

$$r = 0.02$$

Find

Virtual temperature.

Solution

$$\begin{aligned} T_v &= 300(1 + 0.61 \times 0.02) \\ &= 300(1.0122) \\ &= 303.66K \end{aligned}$$

Virtual temperature

$$\boxed{303.7K}$$

Physical Significance

Virtual temperature is used because

Actual temperature alone cannot describe moist air.

It helps in

- Weather forecasting
 - Cloud formation
 - Thunderstorm prediction
 - Atmospheric stability
 - Numerical weather prediction
 - Climate models
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Applications

Weather Forecasting

Humidity changes air density.

Virtual temperature corrects this effect.

Aviation

Aircraft performance depends upon air density.

Virtual temperature helps estimate air density.

Meteorology

Used in

Weather balloons

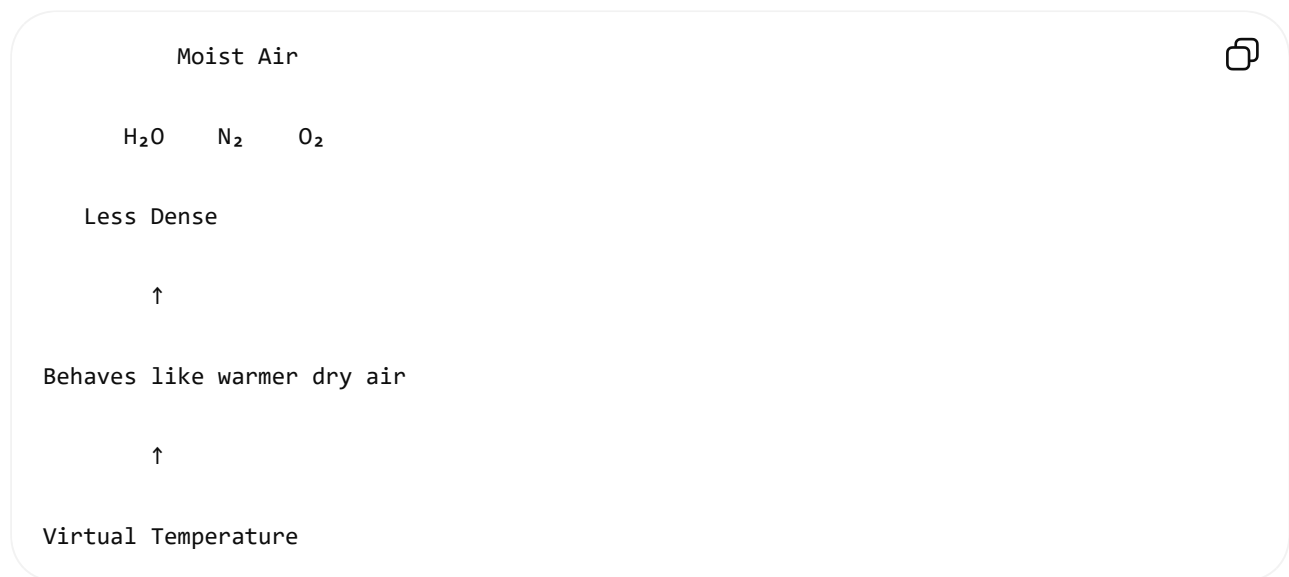
Radiosonde observations

Weather models

Climate Studies

Climate scientists include virtual temperature in atmospheric simulations.

Diagram to Draw on the Board



Important Points for Students

- Moist air is lighter than dry air.
- Water vapour has lower molecular weight.
- Virtual temperature is always equal to or slightly greater than actual temperature.
- Used in meteorology and weather forecasting.

Frequently Asked Questions

Why is moist air lighter?

Because water vapour has molecular weight 18 while dry air averages 29.

Is virtual temperature the actual temperature?

No.

It is an equivalent temperature used for calculations.

Unit

Kelvin (K)

MCQs

1

Water vapour has molecular weight

A) 16

B) 18

C) 29

D) 32

Answer: B

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Virtual temperature is generally

A) Less than actual temperature

B) Equal to actual temperature

C) Greater than or equal to actual temperature

D) Zero

Answer: C

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Virtual temperature is mainly used in

A) Optics

B) Electricity

C) Meteorology

D) Acoustics

Answer: C

University Questions

Short Answer

Define virtual temperature.

Explain why moist air is lighter than dry air.

Long Answer

Explain the concept of virtual temperature with derivation, physical significance, and applications.

Teaching Tip

Start the class by asking, "**Why do people often say humid weather feels heavier, even though moist air is actually less dense?**" This creates curiosity and leads naturally into the concept of virtual temperature. Then connect the discussion to weather forecasting and why meteorologists need a quantity like virtual temperature.

In the next lecture note, I'll prepare **Hydrostatic Equation** in the same detailed style, including the full derivation, force balance on an air parcel, physical interpretation, numerical examples, and applications.

       Sources